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AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

- Problem Statement ID - 26104
- Problem Statement Title - AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- Theme - Blockchain and Cybersecurity
- PS Category - Software
- Team Name - HexaCore

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

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Proposed Solution

- Detects AI Voice Artifacts
- Verifies True Identity
- Flags Scam Words
- Payment/UPI request detection
- Calculates Live Risk Score
- Voice anti-spoofing/deepfake detection

How it address the problem?

- Neutralizes Synthetic Audio Mimicry
- Prevents Caller ID & Identity Impersonation
- Mitigates Social Engineering & Panic Tactics
- Blocks Financial Fraud at the Source
- Eliminates Black-Box Unreliability:

Innovation and Uniqueness

- Tri-Factor Hybrid Security Pipeline- integrats deepfake detection, biometric voice matching and UPI detection
- Phase & Spectral Artifact Mining - finds, extracts, and fixes distortion errors in sound signals so they look and sound clean.
- Sub-Second Low-Latency Edge Execution- Uses quantized, lightweight ML models
- Context-Aware Risk Aggregation

TECHNICAL APPROACH

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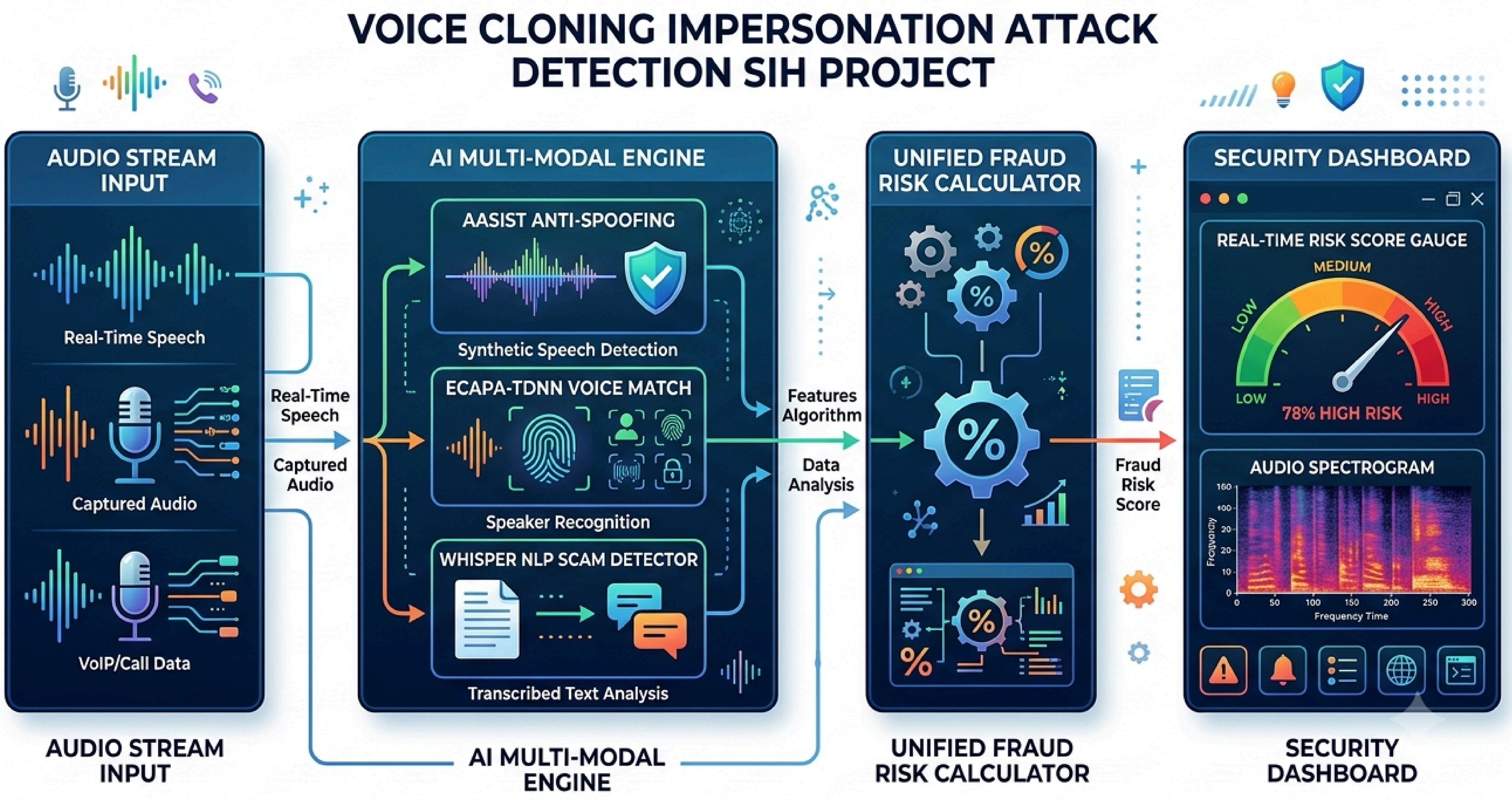
Technologies to be used

Programming Languages: Python (Machine Learning, Signal Processing, Backend API), JavaScript/TypeScript (Frontend Dashboard)

Process for Implementation



FEASIBILITY AND VIABILITY

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Analysis

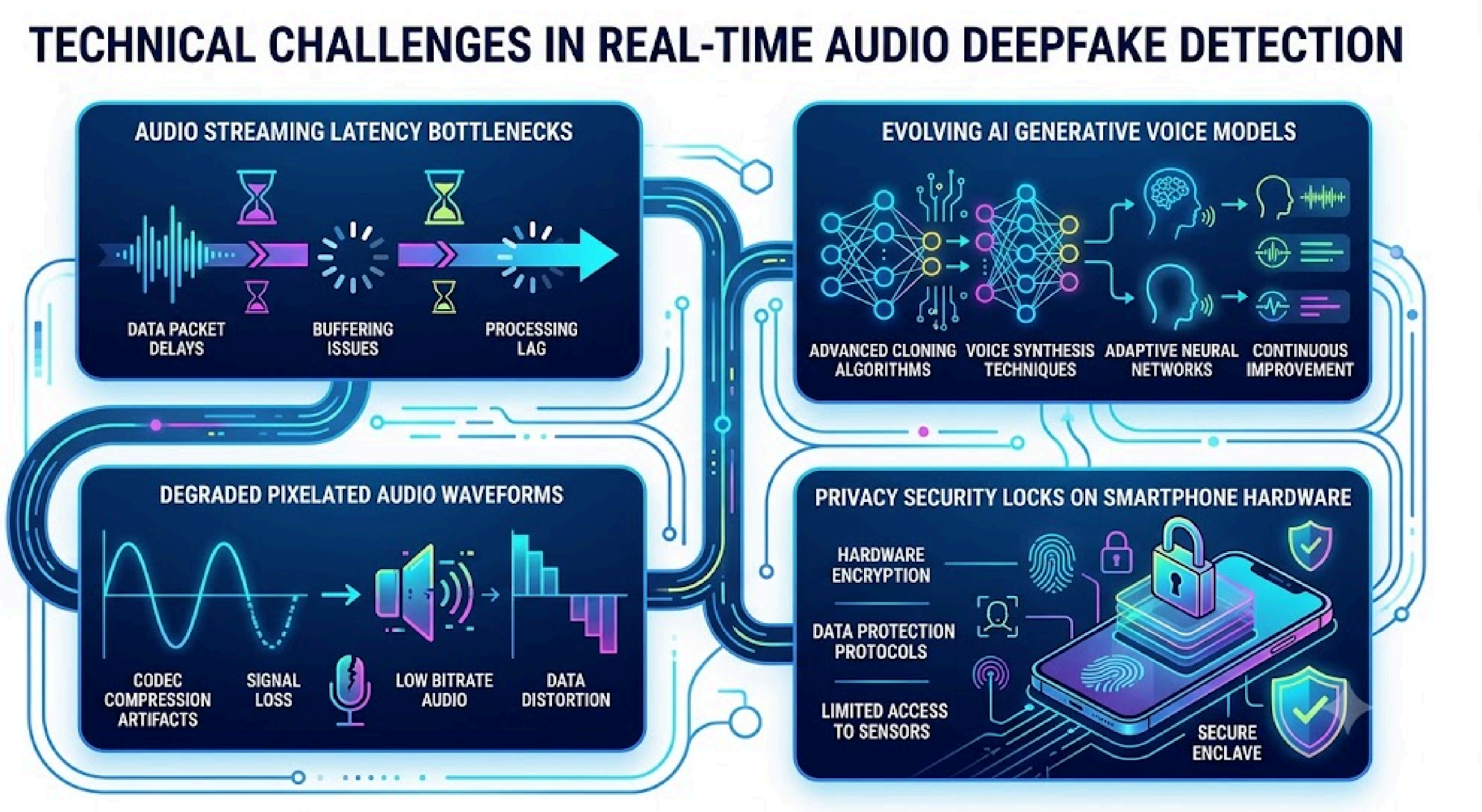
- Model Availability & Proven Architectures
- Feature Extraction Pipeline
- Hackathon Timeline Scope
- Low Hardware Overhead
- Low false-positive rate
- Background noise flaws

Strategies

- Quantisation and Graph Optimization
- Codec Data Augmentation
- Ensemble Feature Extraction
- Continuous Out of Distribution Monitoring
- Zero-Knowledge On-Device Processing

Potential Challenges and Risks

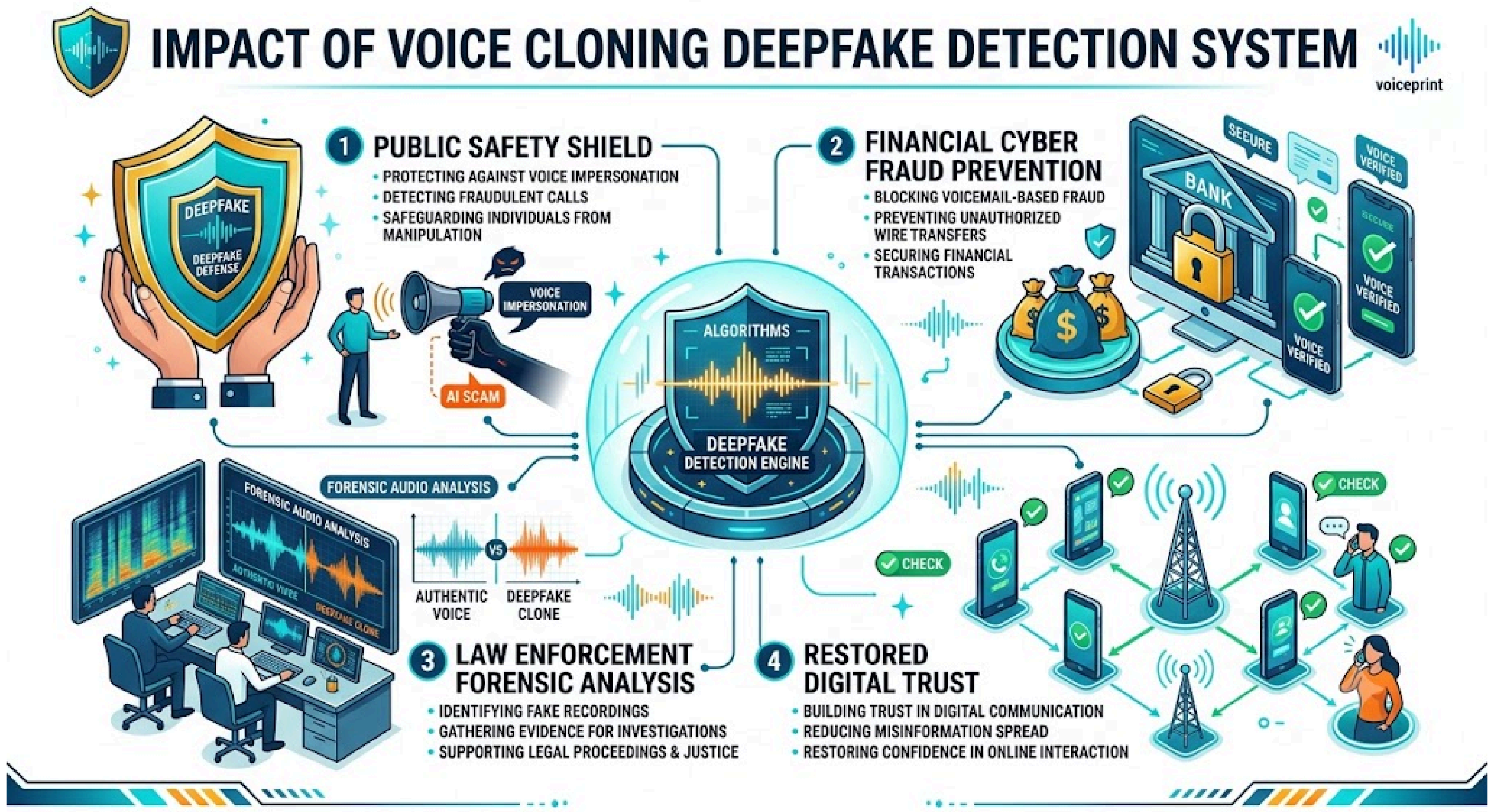
- Latency Bottlenecks in Real-Time Audio Streaming
- Audio Compression and Codec Artifacts
- Rapidly Evolving AI Voice Generators
- Privacy and On-Device Processing Constraints



Imapact And Benefits

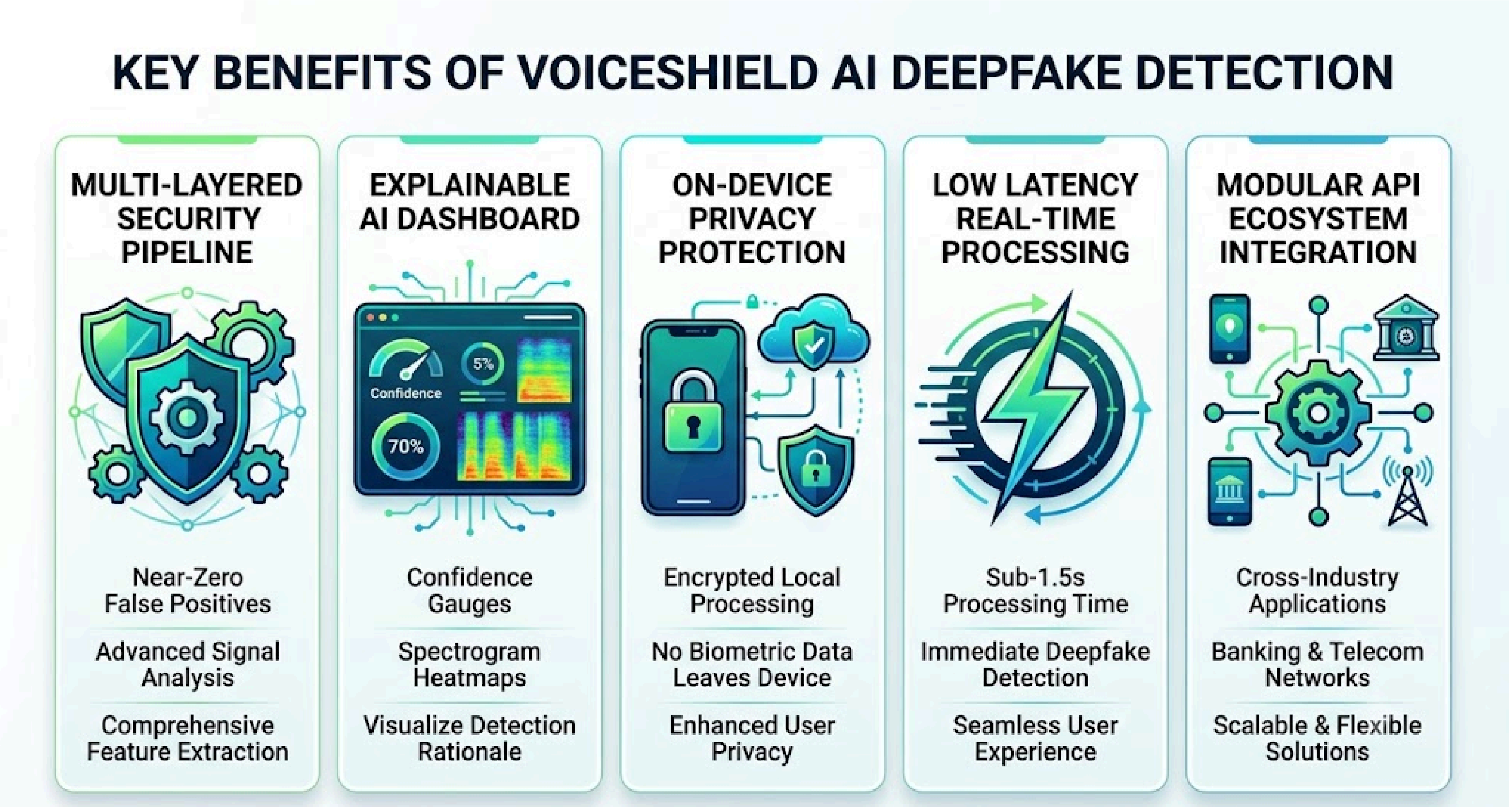
Potential Impact

- Public Safety & Fraud Elimination
- Financial Loss Prevention
- Preservation of Biometric Integrity
- Forensic & Law Enforcement Empowerment
- Reduction in Legal & Dispute Costs



Benefitst

- Multi-Layered Protection
- Explainable Trust & Transparency
- Privacy-Preserving Architecture
- Low-Latency Scalability
- Seamless Ecosystem Integration



Research and References

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  https://arxiv.org/abs/2110.01200
- A Large-Scale Speaker Identification Dataset
  https://www.robots.ox.ac.uk/~vgg/data/voxceleb/
- Robust Speech Recognition via Large-Scale Weak Supervision
  https://www.robots.ox.ac.uk/~vgg/data/voxceleb/